

Madhav Pradeep

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🏢 Professional Experience

Full Stack Developer, Drexon Industries *Apr 2026 – Present*

- Architected geospatial backend (PostgreSQL/PostGIS, Redis) with RBAC, track correlation, and ML anomaly detection
- Built 6 protocol adapters (ASTERIX, MAVLink, NMEA, SNMP, ZMQ) with AES-256-GCM over Redis pub/sub
- Built React.js frontend and WebSocket/REST APIs serving real-time analytics, track history, and alerts

AI-ML Engineer, Novaforge AI *Jan 2026 – Apr 2026*

- Engineered an LLM-powered RAG system over 10K+ documents, delivering context-aware semantic query responses
- Built a FAISS-indexed vector retrieval pipeline; grew client web traffic 10% via ML and SEO

🎓 Education

B.Tech, Mechanical Engineering, IIT Madras *2024 – 28 | 8.80*

Class XII, HSE Kerala – Kuriakose Elias EMS *2024 | 99%*

Class X, CBSE – Sree Narayana Public School *2022 | 94%*

🏆 Scholastic Achievements

- **AIR 2276** JEE Advanced, **AIR 1957** JEE Mains 2024; **INMO 2021** & **NSEA Stage-II** qualifier

⚙️ Technical Experience

Software Lead, Team Abhiyaan *Apr 2025 – Present*

- Deployed SegFormer vision transformer for real-time lane detection, optimizing TensorRT inference to 30+ FPS
- Configured Nav2 and calibrated stereo cameras for visual odometry via URDF transforms in ROS2

Contingent Member, Inter IIT Tech Meet *Oct 2025 – Dec 2025*

- Built a low-latency ROS2-Teensy 4.1 UART interface and modular PCB with 433MHz RF E-stop for an autonomous rover

Languages: Python, C/C++, JavaScript, Node.js, SQL, Bash, Git, Altium, Figma

Backend & DB: React, FastAPI, REST, WebSocket, Docker, PostgreSQL/PostGIS, MongoDB, Redis

ML & Robotics: PyTorch, OpenCV, NumPy, Pandas, TensorRT, ROS2, Nav2, Gazebo

🔧 Projects

Nemetschek x IITM AI Hackathon *2nd Runner-Up, INR 1L*

- Built an HRNet-W32 model with attention to segment floorplans, achieving 0.93 mIoU on test data
- Built a neurosymbolic pipeline converting 2D floorplans to IFC-compliant 3D BIM

Order Flow Imbalance *Self Project*

- Modeled NASDAQ order-flow imbalance, proving linear price impact at 4.1% OOS R^2
- Backtested 384 out-of-sample trades at 67% hit rate, isolating true signal from microprice leakage

Inverted Pendulum on a Cart *ME2400 Project*

- Built an ESP32 inverted-pendulum control system; simulated LQR and PID controllers in Pygame

🏆 Competitions

IGVC 2026, Team Abhiyaan *5th Design, Intl.*

- Secured 5th in the Design Report internationally before a panel of 11 industry judges
- Engineered a 200m+ wireless E-Stop (LoRa, NRF, ESP32) with AES-256-GCM encryption against RF spoofing

QRT Quant Challenge *Top 10, IITM*

- Ranked Top 10 at Qube Research & Technologies' Quant Challenge, advancing to the Phase 2 round
- Built multi-indicator strategies (Ichimoku, RSI, Williams %R); IS Sharpe 2.5, OS Sharpe 1.2+

Reward Rush – RL Competition *Runner-up, INR 1.5k*

- Runner-up among 50+ teams; benchmarked 6 RL algorithms (DQN, SAC, A2C, SARSA, MC) across Gym environments

👥 Leadership

Contingent Captain, IITM Tech Contingent *Mar 2026 – Present*

- Selected to the 3-member core team leading institute-wide technical contingent strategy
- Authored recruitment assignments in ROS2, YOLO, SLAM, and path planning; introduced VLSI and FPGA verticals

Director, Tensors NGO *Mar 2026 – Present*

- Scaled outreach via JEE Mocks, JoSAA Counselling, and Olympiads to 20k+ students across 14 states
- Conducted 8+ workshops for 60+ students at an accredited SOS Village for underprivileged children

Student Mentor, Avanti Fellows *Sep 2024 – Jul 2025*

- Mentored 20+ underprivileged JEE aspirants; built mock-exam analytics to tailor prep and boost scores

★ Extracurriculars

Oratory Club Associate *Oct 2024 – Mar 2025*

- Closed INR 1.2 Lakhs in sponsorships and PR for the IIT Madras Parliamentary Debate, expanding event reach

Quizzes & Elocutions *Multiple 1st Places*

- Won 1st in the KGMOA Medi-IQ health quiz and the Kottayam District Maths Fair talent search
- Bagged 1st in Extempore English at the Sahodaya Kalotsav and SNPS Youth Festival elocutions